Elements: CryoStack — A Sustained Cyberinfrastructure Platform for Ensemble Data Assimilation in Ice-Sheet Science

Principal Investigator: Brian Kyanjo

Georgia Institute of Technology

Submitted in response to: NSF 22-632: Cyberinfrastructure for Sustained Scientific Innovation (CSSI) — Elements

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Project Summary

Overview

This project will grow **CryoStack** into a sustained cyberinfrastructure platform for ensemble-based data assimilation (DA) in ice-sheet and glaciological science. CryoStack, hosted on GHUB, already provides browser-based access to **ICESEE** (an open-source, MPI-parallel ensemble Kalman filter framework) and is being extended with **CryoLauncher**, a browser-mediated HPC/cloud job orchestration layer, and *[fourth component name to be confirmed]*, a verification and validation service. This proposal will harden these into an integrated platform: adding new ice-sheet model couplings (including PISM, in addition to the existing ISSM and Icepack couplings), expanding CryoLauncher's HPC/cloud backend support, and building a program of community activities – workshops, contributor onboarding, and user support – so that the broader glaciology/cryosphere community can adopt ensemble DA workflows without needing to build this infrastructure themselves.

Intellectual Merit

Replace/expand: CryoStack advances both domain science (transient, observation-constrained ice-sheet state and parameter estimation across multiple community models) and cyberinfrastructure practice (a reusable pattern for delivering ensemble DA as a browser-mediated platform service rather than a install-it-yourself library).

Broader Impacts

Replace/expand: by adding PISM support, CryoStack extends ensemble DA capability to a substantially larger existing ice-sheet modeling community than ICESEE alone currently reaches. Community workshops, onboarding documentation, and a no-local-HPC-required entry point broaden participation among students and researchers at resource-limited institutions.

1 Introduction and Motivation

Ensemble-based data assimilation (DA) – particularly the Ensemble Kalman Filter (EnKF) – has emerged as a principled way to fuse ice-sheet model forecasts with observations, jointly constraining model state and parameters without requiring model linearization [Evensen, 2003, Gillet-Chaulet, 2020, Choi et al., 2025]. The PI's team previously developed **ICESEE**, an open-source, MPI-parallel stochastic EnKF framework demonstrated with ISSM [Larour et al., 2012] and Icepack [Shapero et al., 2021], achieving near-linear scaling to 320 MPI ranks on Georgia Tech's PACE cluster [Kyanjo et al., 2026a].

That work established the algorithmic core. The barrier that remains is access: running ensemble DA still requires a Python/MPI environment, HPC job-submission expertise (SSH/SLURM), and prior fluency in ensemble DA concepts – excluding much of the glaciology/ cryosphere community this capability is meant to serve, including researchers at institutions without HPC access or dedicated software engineering support.

This proposal addresses that barrier by growing **CryoStack** into a sustained cyberinfrastructure *platform*, rather than treating ICESEE as a standalone library with a documentation site attached. CryoStack, GHUB-hosted, becomes the single point of entry; ICESEE (the DA engine), **CryoLauncher** (browser-mediated HPC/cloud orchestration), and a *[fourth component name to be confirmed]* verification and validation service become pluggable backend services it coordinates. Concretely, this proposal will: (1) add new ice-sheet model couplings to the platform, notably **PISM** (Parallel Ice Sheet Model), a widely used community model with a substantially larger existing user base than ISSM or Icepack alone, plus additional couplings as capacity allows; (2) extend CryoLauncher's HPC/cloud backend support; and (3) establish a sustained program of community-building activities – workshops, contributor onboarding, and user support channels – so CryoStack grows as a community-maintained platform rather than a single-lab tool.

2 Results from Prior NSF Support

The ICESEE project has received prior NSF support under Award No. 2235920 (NSF Office of Polar Programs, GEO/OPP), which funded the core development of the ICESEE DA engine now serving as CryoStack's first backend service.

[Confirm and complete: award title, PI(s) of record, award amount, and period of performance.]

Intellectual Merit outcomes: development of ICESEE (v1.0.0), demonstrated coupling with ISSM and Icepack, a hybrid ensemble assimilation–inversion strategy for weakly observed parameters (e.g., basal friction), and near-linear MPI scaling to 320 ranks on PACE Phoenix [Kyanjo et al., 2026a].

Broader Impacts outcomes: open-source release (BSD 2-Clause) with an archived, citable Zenodo release [Kyanjo et al., 2026b] and a continuous integration pipeline validating every commit against a Lorenz-96 benchmark. *[Add: training, workshop, or student outcomes tied to this award.]*

Relevance to this proposal: Award No. 2235920 built the engine; this CSSI Elements proposal builds the platform around it – turning a validated research library into sustained, community-facing cyberinfrastructure.

3 Project Motivation and Impact

3.1 Science-Driven

Replace/expand: adding PISM support is the central science-driven argument for this proposal – it extends ensemble DA capability to PISM's existing community (a substantially larger and more diverse user base than ISSM/Icepack combined), enabling observation- constrained state and parameter estimation for a

much broader set of glaciological research questions than ICESEE alone currently reaches. Discuss education/workforce development plans here as well.

3.2 Innovation

Replace/expand: the innovation here is as much in cyberinfrastructure engineering as in DA methodology – a platform architecture that cleanly separates (a) a stable, versioned DA engine API (ICESEE) that new model couplings implement against, (b) a job-orchestration layer (CryoLauncher) that abstracts away HPC/cloud differences, and (c) an automated verification/validation service that gates new model couplings before they reach users. This plug-in architecture is what makes adding PISM (and future models) a bounded, repeatable engineering task rather than a bespoke integration each time.

4 Cyberinfrastructure Plans

4.1 Project Plans, and System and Process Architecture

Replace/expand with the platform architecture: CryoStack (GHUB-hosted access layer: documentation, tutorials, launch UI) sits atop a pluggable backend-services layer comprising ICESEE (DA engine), CryoLauncher (HPC/cloud job orchestration, submitting to PACE and AWS), and *[fourth component]* (verification/validation). Describe: (a) the plug-in contract new model couplings (e.g., PISM) must implement to register with ICESEE and become launchable through CryoLauncher; (b) the engineering process for design, development, documentation, testing, validation, and release, including how the verification/validation service gates new couplings before they are exposed to users; (c) the deployment and end-user acceptance/evaluation plan; (d) security/privacy and credential handling for CryoLauncher's HPC/cloud job submission, plus provenance, transparency, reproducibility, and usability; and (e) how the platform will adapt to new technologies and community requirements. **Because this is an Elements proposal, also include your project management plan here** (PI/Co-PI roles and division of labor across the DA-engine, orchestration, V&V, and community-building work streams; milestones; decision-making process) rather than as a separate supplementary document.

4.2 Building on Existing, Recognized Capabilities

Replace/expand: CryoStack builds on GHUB (`https://theghub.org`) for browser-based deployment, Georgia Tech's PACE HPC cluster, and (for CryoLauncher's cloud backend) AWS. Discuss any relationship to the ACCESS program and to the PISM community's existing infrastructure and documentation, since integrating with an established community model means coordinating with, not duplicating, its existing user base and support channels.

4.3 Close Collaborations Among Stakeholders

Replace/expand: describe how the PI (leading DA-engine and platform integration work) and Co-PI (leading science-application and community engagement) will divide and coordinate work, and how outside stakeholders – e.g., PISM developers/users, GHUB platform staff – will be engaged as new couplings are added.

5 Measurable Outcomes

5.1 Deliverables

Replace/expand: (1) CryoStack platform releases integrating PISM (and additional models as capacity allows) as launchable options through CryoLauncher; (2) expanded CryoLauncher HPC/cloud backend support; (3) an automated verification/validation service gating new model couplings; (4) a sustained community-building program (see below). Cross-reference the Delivery Mechanism and Community Usage Metrics

supplementary document for full detail.

5.2 Community Building and Growth

Replace/expand with the specific activities planned to grow CryoStack as a community-maintained platform rather than a single-lab tool:

- **Workshops/tutorials:** recurring sessions (virtual and/or co-located with community events such as AGU, IGS, or the PISM community's own meetings) teaching ensemble DA concepts and hands-on CryoStack use.
- **Contributor onboarding:** a documented plug-in contract and contribution guide so external developers can add new model couplings without core-team involvement; "good first issue" labeling; a public roadmap.
- **User support:** a persistent discussion/support channel (e.g., GitHub Discussions) with a stated response-time norm.
- **Governance:** a lightweight, documented decision- making process for accepting new couplings and community contributions, sized appropriately for a small core team but signaling a credible path to broader maintainership.

5.3 Sustained and Sustainable Impacts

Replace/expand: how these community-building activities create a pathway toward reduced single-lab dependency (e.g., external contributors maintaining specific model couplings) and toward a future Transition to Sustainability proposal once adoption metrics support it.

5.4 Metrics

Replace/expand with a brief summary of year-by-year metrics spanning both technical delivery and community growth (full detail in the Delivery Mechanism and Community Usage Metrics supplementary document).

6 Broader Impacts

Replace/expand as an explicit, standalone Broader Impacts statement: broadening participation via PISM's larger/more diverse user base; training and workforce development through workshops; accessibility gains from a browser-based, no-HPC-required entry point; a governance model designed to outlive any single funded position.

7 Alignment with NSF Directorates/Divisions

Replace with an explicit statement of which participating directorate(s)/division(s) (e.g., CISE/OAC, GEO/OPP, GEO/EAR) this proposal is most aligned with – note the existing relationship with GEO/OPP via Award No. 2235920.

8 Timeline

Replace with a 3-year timeline spanning platform engineering and community-building milestones. A simple table works well.

References Cited

References

Y. Choi, A. Petty, D. Felikson, and J. Poterjoy. Estimation of the state and parameters in ice sheet model using an ensemble kalman filter and observing system simulation experiments. *EGUsphere*, pages 1–25, 2025. doi: 10.5194/egusphere-2025-301.

Geir Evensen. The ensemble kalman filter: theoretical formulation and practical implementation. *Ocean Dynamics*, 53:343–367, 2003. doi: 10.1007/s10236-003-0036-9.

F. Gillet-Chaulet. Assimilation of surface observations in a transient marine ice sheet model using an ensemble kalman filter. *The Cryosphere*, 14:811–832, 2020. doi: 10.5194/tc-14-811-2020.

Brian Kyanjo, Talea L. Mayo, and Alexander A. Robel. The ice sheet state and parameter estimator (icesee) library (v1.0.0): Ensemble kalman filtering for ice sheet models, 2026a. Manuscript submitted for publication.

Brian Kyanjo, Alexander Robel, and Talea L. Mayo. Icesee (v1.0.0): Ice sheet state and parameter estimator. Zenodo, 2026b.

E. Larour, H. Seroussi, M. Morlighem, and E. Rignot. Continental scale, high order, high spatial resolution, ice sheet modeling using the ice sheet system model. *Journal of Geophysical Research: Earth Surface*, 117:F01022, 2012. doi: 10.1029/2011JF002140.

D. R. Shapero, J. A. Badgeley, A. O. Hoffman, and I. R. Joughin. icepack: a new glacier flow modeling package in python, version 1.0. *Geoscientific Model Development*, 14:4593–4616, 2021. doi: 10.5194/gmd-14-4593-2021.

Biographical Sketch: Brian Kyanjo

NOTE: NSF requires biographical sketches to be generated through SciENcv (`https://www.ncbi.nlm.nih.gov/sciencv/`) and submitted as PDF attachments. The outline below mirrors the required structure for internal drafting purposes only.

(a) Professional Preparation

Replace with institution, major, degree, and year for each degree.

(b) Appointments

Replace with a reverse-chronological list of positions held.

(c) Products

Products Most Closely Related to the Proposed Project

1. Replace with citation.
2. Replace with citation.

Other Significant Products

1. Replace with citation.
2. Replace with citation.

(d) Synergistic Activities

1. Replace with activity (e.g., service, mentoring, outreach).
2. Replace with activity.

Budget Justification

Senior Personnel

PI (Brian Kyanjo), Georgia Institute of Technology: 60% effort for 3 years, leading platform integration, the ICESEE DA engine, and new model-coupling development (including PISM). *[Confirm actual base salary – placeholder used in the companion worksheet: $85,000/yr.]*

Co-PI (Alexander A. Robel), Georgia Institute of Technology: one summer month per year for 3 years, leading science-application oversight and community/stakeholder engagement (including PISM community outreach). *[Confirm actual 9-month academic-year salary – placeholder used in the companion worksheet: $130,000/yr.]*

[Confirm: is Talea L. Mayo (Emory University) included as Senior/Key Personnel with funded effort? If so, this becomes a multi-institution collaborative proposal requiring a subaward budget at Emory, which is not yet reflected in the companion worksheet.]

Other Personnel

Replace with justification for any postdocs, graduate/undergraduate students, or research software engineer (RSE) effort supporting new model-coupling development or the verification/validation service – not currently budgeted in the placeholder worksheet (set to $0/yr); add if planned.

Equipment

Not currently anticipated; PACE and AWS compute needs are covered under Other Direct Costs (cloud compute) below rather than as capital equipment. Replace if any single item over $5,000 is needed.

Travel

CSSI-required annual PI meeting: 1 trip/year near NSF (Alexandria, VA) for 3 years, placeholder $1,800/trip. **Other travel:** conference presentation and/or PISM-community engagement travel, placeholder $3,000/year. Replace both with actual anticipated costs.

Participant Support Costs

Workshop/tutorial participant support (travel and/or stipends for external attendees), placeholder $6,000 per workshop, 1 workshop/year for 3 years. These funds are excluded from the indirect-cost base and cannot be redirected to other categories without NSF approval. Replace with the actual number and scale of planned workshops (see Community Building and Growth in the Project Description).

Other Direct Costs

Materials and supplies: placeholder $1,500/year. **Publication/open-access costs:** placeholder $1,500/year. **Cloud compute (AWS):** placeholder $4,000/year, supporting CryoLauncher cloud-backend testing and demonstration workloads distinct from PACE HPC allocations covered under Facilities. Replace all three with actual anticipated costs.

Indirect Costs

Georgia Tech's federally negotiated indirect cost (F&A) rate, applied to Modified Total Direct Costs (MTDC) – i.e., total direct costs excluding Participant Support Costs. *[Confirm GT's current negotiated rate; placeholder of 57% used in the companion worksheet.]*

Budget Cap Check

CSSI Elements caps total budget (all years, direct + indirect) at $600,000. Using the placeholder figures above, the companion worksheet computes a 3-year total of approximately $486,000 – within the cap, but with limited headroom if actual salary figures are higher than the placeholders, if Emory becomes a funded subaward, or if RSE/student support is added. Recalculate with real numbers before finalizing the Research.gov budget forms.

Facilities, Equipment and Other Resources

Facilities

Replace with a description of relevant facilities – e.g., Georgia Tech's PACE (Partnership for an Advanced Computing Environment) HPC cluster used for ICESEE/CryoLauncher compute jobs, departmental server infrastructure hosting CryoStack's GHUB deployment, and any lab space.

Equipment

Replace with a description of major equipment already available (distinct from equipment requested in the budget) – e.g., existing compute allocations, storage systems for model output/observational data.

Other Resources

Replace with other resources such as GHUB platform infrastructure (`https://theghub.org`) supporting CryoStack/CryoLauncher deployment, institutional support, or external collaborations (e.g., access to ISSM/Icepack model teams, observational data providers).

Data Management and Sharing Plan

Types of Data Produced

Replace with a description of the data, software, and other materials generated – e.g., ICESEE source code and releases; CryoStack documentation/tutorial content; CryoLauncher orchestration code; synthetic benchmark datasets (e.g., Lorenz-96 test cases); any observational/model output data used in ice-sheet DA experiments.

Data Formats and Standards

Replace with the formats/standards used – e.g., HDF5/NetCDF for scientific data pipelines (as ICESEE already uses), open-source code in Python, Jupyter Book/Markdown for documentation.

Data Access and Sharing Policies

Replace with how and when code/data will be shared. ICESEE is already released under the BSD 2-Clause License on GitHub (`https://github.com/ICESEE-project/ICESEE`); state whether CryoStack and CryoLauncher will retain their current licenses (note: CryoLauncher is currently MIT-licensed) or be harmonized, and justify the choice(s).

Archiving and Preservation

Replace/confirm: ICESEE already maintains a citable archival release on Zenodo (DOI: 10.5281/zenodo.21796612) alongside active GitHub development, consistent with FAIR principles. State the plan for continuing this practice (e.g., a new Zenodo DOI minted at each tagged release) for CryoStack and CryoLauncher as well.

Roles and Responsibilities

Replace with who is responsible for managing and sharing data/code during and after the award period (e.g., the PI and any funded CI Professional maintaining the ICESEE/CryoLauncher repositories).

Delivery Mechanism and Community Usage Metrics

Deliverables: How will services and capabilities be delivered?

- **CryoStack** (platform): GHUB-hosted access layer (`https://cryostack.eas.gatech.edu`) – documentation, tutorials, and the launch UI coordinating the backend services below.
- **ICESEE** (DA engine): versioned, open-source releases on GitHub (`https://github.com/ICESEE-project/ICESEE`), installable via pip/conda/containers/Spack, with a documented plug-in contract for new model couplings (target: PISM in addition to existing ISSM/Icepack couplings).
- **CryoLauncher** (orchestration): browser-based job submission to HPC (PACE) and cloud (AWS) backends, removing the need for users to manage SSH/SLURM directly.
- *[Fourth component – name TBD]* (verification/ validation): automated benchmarking that gates new model couplings before they reach users.

Metrics: How will success be measured?

Replace with concrete, quantifiable, year-by-year metrics spanning both technical delivery and community growth:

- **Development metrics:** tagged releases per year; existing GitHub Actions CI already validates every commit against a Lorenz-96 benchmark – extend to cover new couplings; number of supported model couplings (currently ISSM, Icepack, Lorenz-96, flowline; target: + PISM).
- **Delivery metrics:** GHUB/CryoStack tool launches per quarter; CryoLauncher HPC/cloud job submissions per quarter (ICESEE has already demonstrated near-linear scaling to 320 MPI ranks on PACE Phoenix, a relevant baseline); documentation page views.
- **Adoption/usage metrics:** GitHub stars/forks/unique clones across CryoStack-project repositories; number of external research groups using the platform (track PISM- community adoption specifically, given its larger existing user base); citations to the ICESEE software paper [Kyanjo et al., 2026a].
- **Community-growth metrics:** workshop/tutorial attendance per year; number of external contributors submitting a model coupling or code contribution; support- channel response time; contribution-guide/onboarding completion rate.

For each metric, state the Year 1 / Year 2 / Year 3 numeric target and how it will be collected (GitHub API/Insights, GHUB usage analytics, workshop registration counts, support-channel logs). Replace the placeholder table below with real targets.

Collection Methodology

Replace with the mechanisms used to collect the above (GitHub Insights/API, GHUB platform analytics, workshop registration, opt-in user registration at first CryoLauncher use, annual user survey).

Project Personnel and Partner Organizations

1. Brian Kyanjo; Georgia Institute of Technology; PI
2. Alexander A. Robel; Georgia Institute of Technology; Co-PI

irm role/inclusion: Talea L. Mayo; Emory University; [Senior/Key Personnel or Unfunded Collaborator – Dr. Mayo co-authored the ICESEE software paper and advised on data-assimilation methodology; confirm whether she should be listed as Senior Personnel (with a biosketch) or as an Unfunded Collaborator with a Letter of Collaboration]

CI Professional Mentoring and/or Professional Development Plan

Include this document only if the project funds CI Professional personnel (e.g., a research software engineer maintaining ICESEE or CryoLauncher). This plan is evaluated under the Broader Impacts review criterion.

CI Professionals Supported by This Project

Replace with a list of the CI Professional role(s) funded (e.g., "one part-time research software engineer maintaining the CryoLauncher HPC/cloud orchestration layer").

Mentoring and Professional Development Activities

Replace with the specific activities planned, such as:

- Career counseling and pathway guidance.
- Training and support in preparing grant proposals, publications, and presentations (e.g., co-authoring the ICESEE software paper or CI-focused workshop submissions).
- Guidance on collaborating with researchers across science/engineering disciplines and diverse backgrounds.
- Opportunities for professional training (e.g., HPC/cloud certifications, software engineering best-practices training) and career advancement.
- Training in responsible professional practices (e.g., secure credential handling in CryoLauncher's SSH/SLURM backend, reproducibility standards for ICESEE releases).